

JavaScript Tutorials

HTML Tutorials: [Click Here](#)

CSS Tutorials: [Click Here](#)

JavaScript Logic Building Tutorials: [Click Here](#)

GitHub Notes: <https://github.com/muhammadfarhandeveloper/MyNotes>

Programming with JavaScript

Introduction to JavaScript:

- JavaScript is a Scripting Language
- Many used in web pages to improve design, validate forms, detect browsers, and create cookies.
- By default, all web browsers have in built JavaScript support.
- JavaScript is the most popular programming language in the world and that makes it a programmer's great choice. Once you learnt JavaScript, it helps you developing great front-end as well as back-end software's using different JavaScript based frameworks like jQuery, Node.JS etc.

There are many useful **JavaScript frameworks** and libraries available:

- Angular ----- **framework**
- React ----- **library**
- jQuery ----- **library**
- Vue.js ----- **library**
- Node.js ----- **environment or backend**

Website VS Web Application

Server Side VS Client Side Script

How Use JS?

JavaScript object document?

`document.write("Hey");`

how to comment?

Implementation of HTML.

Variables : var , let , const ... difference between.

Data types. String, number, Boolean, Array, object, function, null, undefined.

Chrome console.

Global Variable VS local Variables.

Arithmetic Operator: + , - , * , / , ** , % , ++ , --

Assignment Operator : = , += , -= , *= , /= , % =

Comparison Operator : == , === , != , !== , > , < , >= , <=

If statement in details.

Logical operator. It is use in if statement. && , || , !

Concatenation

Template String.

Alert.

Confirm box

Prompt

Prompt with parseInt();

Class Assignment: make marksheet with parseInt();

Function with recursion functions.

All Loops: For Loop, While Loop, Do while Loop, For of Loop, For In Loop

Assignment: find odd, even numbers using Loop.

Create this using table, loop and array

1	farhan	20
2	ali	25
3	rehman	36
5	Ghaffar	20
6	Ghaffar	20
7	ali	25
8	Rehman baba	40

JavaScript Events

What is an Event?

Events are fired inside the browser window, and tend to be attached to a specific item that resides in it. This might be a single element, a set of elements, the HTML document loaded in the current tab, or the entire browser window.

Event using EventListener?

objects that can fire events have an `addEventListener()` method, and this is the recommended mechanism for adding event listeners.

Examples:

```
button.addEventListener("click", () => {  
  console.log("Clicked");  
});
```

JavaScript Event Categories:

1. Mouse Events

Event	Description
click	Mouse click
dblclick	Double click
mousedown	Mouse button press
mouseup	Mouse button release
mousemove	Mouse move
mouseenter	Cursor enters element
mouseleave	Cursor leaves element
contextmenu	Right click

2. Keyboard Events

Event	Description
keydown	Key pressed
keyup	Key released

3. Form Events

Event	Description
submit	Form submit
input	Value changes while typing
change	Value changed and focus lost
focus	Input focused
blur	Input loses focus
reset	Form reset

4. Windows Events

Event	Description
load	Page loaded
resize	Window resized
scroll	Page scroll
unload	Page closing

DOM (Document Object Model) Scripting

When writing web pages and apps, one of the most common things you'll want to do is change the document structure in some way. This is usually done by manipulating the Document Object Model (DOM) via a set of built-in browser APIs for controlling HTML and styling information.

MDN Reference: https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Scripting/DOM_scripting

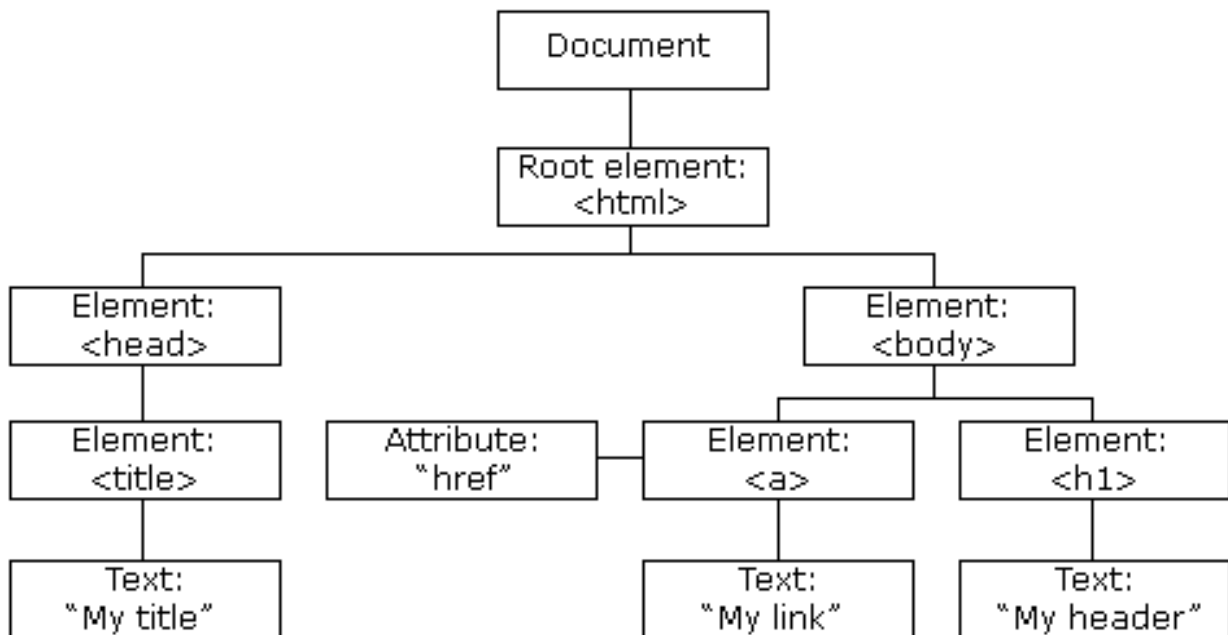


Document Object Model:

The document currently loaded in each one of your browser tabs is represented by a DOM. This is a "tree structure" representation created by the browser that enables the HTML structure to be easily accessed by programming languages — for example the browser itself uses it to apply styling and other information to the correct elements as it renders a page, and developers like you can manipulate the DOM with JavaScript after the page has been rendered.

- JavaScript can change all the HTML elements in the page
- JavaScript can change all the HTML attributes in the page

- JavaScript can change all the CSS styles in the page
- JavaScript can remove existing HTML elements and attributes



HTML DOM Target Methods:

- document.getElementById
- document.querySelector("a")
- document.querySelectorAll(".class")
- document.getElementsbyClassName
- document.getElementsbyTagName

use however you want.

Get with DOM

- | | |
|----------------|---------------------------------|
| • innerText | important |
| • innerHTML | important |
| • textContent | important |
| • getAttribute | for find attribute value |
| • Attributes | for find all Attributes. |

For Set:

- innerText
- innerHTML
- textContent
- Setattribute
- Attributes **use in array syntax**
- Removeattributes.

DOM Styling In javascript

- style
- className **for add class use with =**
- classList **return in array. Add()**

```
document.getElementById("heading").style.backgroundColor = "red";
```

Javascript Add Event Listener Method:

It is use for events.

Syntax:

```
document.getElementById("head").addEventListener("event",functionname);
```

Example:

```
document.getElementById("h1").addEventListener("click",function(){
this.style.backgroundColor = "red";
})
```

Traversal Method in JavaScript:

- parentNode
- ParentElement
- Children
- childNodes
- firstChild
- firstElementChild
- lastChild
- lastElementChild
- nextElementSibling
- nextSibling
- previousElementSibling
- previousSibling

This All use in Practical.

DOM Create Methods:

- createElement
- createTextNode
- createComment

Example:

```
console.log(document.createElement("p"))
console.log(document.createTextNode("Hey i am farhan"))
console.log(document.createComment("end of website"))
```

DOM Append Method:

- appendChild or append
- insertBefore

appendChild:

Example:

```
console.log(document.getElementById("para").append("HEy jf is append text"))
```

Big Example:

```
var element = document.createElement("h1")
var node = document.createTextNode("Hey This Awesome place");
element.append(node)

document.getElementById("para").append(element)
```

insertbefore

it is like appendchild but it is add child in start.

How to get value from input:

Array Methods

Sort

Reverse

Push to add value

Pop to remove value

indexOf .. it is find to data in array and return index num. if don't so return -1;

includes method for name but it is return true or false.

Call back functions:

Find : to find a first index number for find start values.

FindIndex: same as a find method. a first index number for find index number.

Filter to find values in array and create another array.

Syntax;

```
var age = [20,50,20,1,4,14]
console.log(age)

var b = array.filter(function(nam){
    return nam>=18
})

console.log(b)
```

For each Loop:

Just like another but It is very easy as compare to for loop.

Syntax:

```
var array = ["farhan","ali","arif","rehman"]

array.forEach(function(v1){
    document.write(v1 + " <br>")
})
```

```
Array.forEach(function(val,index){  
Document.write(val + index);  
})
```

For In LOOP:

It is use only for object.

```
var obj = {id :1 , name: "farhan",age : 20}  
for (const key in obj) {  
  console.log(obj[key])  
}
```

Map Method In JS:

It is use for creating gallery or other things.

```
array.map(function(val,index){  
  console.log(val + index)  
})
```

Date Use in JS:

```
var now = new Date();  
console.log(now.toString());  
console.log(now.toTimeString());
```

we can fix the date.

```
var now = new Date("June 19 2021");  
console.log(now.toString());
```

Set Interval & Clear Interval:

```
setInterval(function(){  
    document.write("farhan");  
}, 1000);
```

Set Clear is use for clear the interval but in any condition:

Syntax:

SetClear(setInterval variable name);

Set Timeout & set clear:

It is running like a set interval but it is run only one time:

Syntax:

```
setTimeout(function(){  
    console.log("hello")  
},5000)
```

ClearTimeout:

Syntax:

clearTimeout(setInterval variable name);

Example:

```
var a = setTimeout(function(){  
    console.log("hello")  
},5000)  
  
function stopanim(){  
    clearTimeout(a)  
  
}
```

BOM (Browser Object Model)

- Get Width & Height of Browser Window
- Open & Close Browser Window
- Move & Resize Browser Window
- Scroll to Browser Window
- Get URL, Hostname, Protocol of Browser Window
- Get History of Browser Window

Window Height & Width:

For Method to Get browser (window) height & width.

- innerHeight
- innerWidth
- outerHeight

- `outerWidth`

It is also work with events. like `onscroll` , `onresize`.

Window Open & Close Method

Open:

`Window.open(URL,name,specs)`

- URL: website link with protocol.
- Name or Target: give target it is great. `_blank` , `_self` etc.
- Spec: Width, height, left, top

`Window.close()`

Location Object:

Describe location object.

And use in `console.log`.

- `hash`
- `host`
- `hostname`
- `href`
- `origin`
- `pathname`
- `port`
- `protocol`
- `search`

Location Methods.

`Location.reload()`

`Location.assign("url")`

`Location.replace("")`

Replace like assign but it destroy history.

History Object (length):

`History.back`

`History.forward`

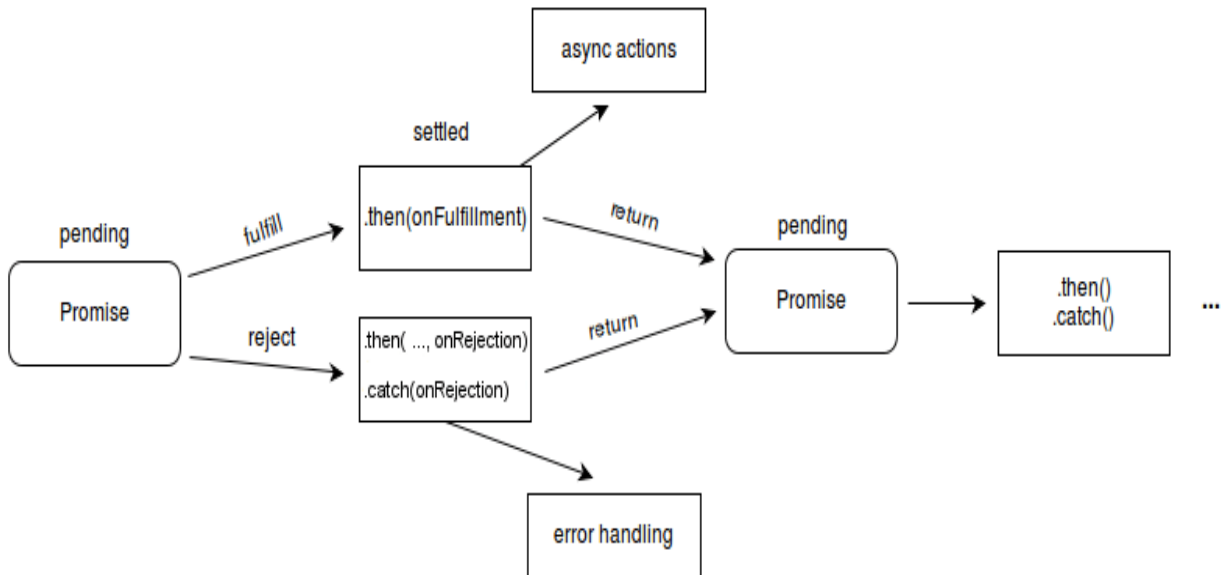
`History.go` it is use for both like `history.go(1)` or `(-1)`

`window.scrollTo` with use on event.

Fetch the API:

What is Promise?

The Promise object **represents the eventual completion (or failure) of an asynchronous operation and its resulting value.**



Fetch in method:

The `fetch()` method in JavaScript is **used to request to the server and load the information in the webpages**. The request can be of any APIs that returns the data of the format JSON or XML. This method returns a promise. Syntax: `fetch( url, options )`.

```
fetch('https://fakestoreapi.com/products')
  .then(res=>res.json())
  .then(json=>console.log(json))
```

Simple with normal function:

```
fetch("https://jsonplaceholder.typicode.com/users").then(function(res){
    return res.json();
})
.then(function(resp){
    console.log(resp)
})
```

Another Example:

```
fetch("https://jsonplaceholder.typicode.com/users").then(function(res){
    return res.json();
})
```



```
).then(function(resp){
  console.log(respa)
  for(var i=0;i<resp.length;i++){
    document.write(resp[i].id +") " + resp[i].name + " ---- "+
resp[i].username + "<br>" );
  }
}
).catch(function(error){console.error("You have a error")})
```